

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)
M.Tech Examinations
Assessment - III - (Autumn 2022)

Course Name: Data Structures and Algorithms with C

Course Code: AM607D

Date: 23 - 11 - 2022

Max Marks: 10

Duration: 01 Hr

Note:

1. Marks are indicated against each question. Write clearly and legibly. All diagrams (if any) to be neatly labeled. Attempt all questions. Calculators are allowed.
2. Before attempting please ensure you have the correct question paper.
3. Programming questions may be attempted in C or Python

Q1	Write a program to demonstrate merge sort. You may assume an array with eight elements.	[2.5]
Q2	Explain Merge Sort with the help of an example.	[2.5]
Q3	Write a program to delete an entry in a Binary Search Tree.	[2.5]
Q4	a) There are 10, 15, 13, 44 nodes in 4 different trees. Which of them could have formed a full binary tree? b) Traverse the binary tree using Inorder, Preorder, and Postorder traversals: <pre> graph TD A((A)) --- B((B)) A --- C((C)) B --- D((D)) B --- E((E)) D --- H((H)) C --- F((F)) C --- G((G)) G --- I((I)) G --- J((J)) </pre>	[1.0]
		[1.5]

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ALL THE BEST

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